The osculating mean of the polynomial–logarithmic curve

A solution of MathDB 327618

Research notes in Principia Math In Progress

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1 Task 1: the problem and its solution

Fix $n \geq 2$ and distinct positive numbers $a_1, \dots, a_n$. For

$$\alpha(t) = (t, t^2, \dots, t^{n-1}, \log t), \qquad t > 0,$$

let O_a be the affine hyperplane through $\alpha(a)$ whose direction contains $\alpha'(a), \dots, \alpha^{(n-1)}(a)$. If their intersection is $P = (i_1, \dots, i_n)$, the last coordinate mean is $M_n = \exp(i_n)$. Horwitz's final conjecture in [1] asserts that this equals the multivariate identric mean I_Z defined below. The MathDB title calls this a logarithmic mean conjecture; the formula to be identified is identric.

Put $N = n - 1$, $H_N = \sum_{k=1}^{N} 1/k$, and

$$w_i = \frac{a_i^N}{\prod_{k \neq i}(a_i - a_k)}.$$

Theorem 1. *The hyperplanes $O_{a_1}, \dots, O_{a_n}$ have a unique intersection, and*

$$i_n = \sum_{i=1}^{n} w_i \log a_i - H_N, \qquad M_n = \exp\left(\sum_{i=1}^{n} w_i \log a_i - H_N\right) = I_Z(a_1, \dots, a_n).$$

Thus the source's conjecture holds in every dimension $n \geq 2$.

The alternating Vandermonde formula in the source is exactly this expression: if $V(a) = \prod_{j<i}(a_i - a_j)$ and V_i is the same Vandermonde product after deleting a_i, then

$$\frac{(-1)^{n+i} V_i(a)}{V(a)} = \frac{1}{\prod_{k \neq i}(a_i - a_k)}.$$

This identity holds in the displayed input ordering, without a sorting assumption.

2 Osculating hyperplanes from Taylor polynomials

Lemma 1. *For $a > 0$ the hyperplane O_a has equation*

$$z = \log a - H_N + \sum_{j=1}^{N} \frac{(-1)^{j-1}}{j} \binom{N}{j} a^{-j} x_j.$$

Proof. The first N coordinates of the vectors $\alpha'(a), \dots, \alpha^{(N)}(a)$ form a triangular $N \times N$ matrix with diagonal $1!, 2!, \dots, N!$. They are independent. Hence an osculating hyperplane exists uniquely and can be expressed as a graph in z.

The degree-N Taylor polynomial of $\log t$ about a is

$$Q_a(t) = \log a + \sum_{k=1}^{N} \frac{(-1)^{k-1}}{ka^k}(t-a)^k.$$

Its constant coefficient is $\log a - H_N$. For $1 \le j \le N$, its coefficient of t^j is

$$\begin{aligned}\frac{(-1)^{j-1}}{a^j} \sum_{k=j}^{N} \frac{1}{k}\binom{k}{j} &= \frac{(-1)^{j-1}}{ja^j} \sum_{k=j}^{N} \binom{k-1}{j-1} \\ &= \frac{(-1)^{j-1}}{ja^j}\binom{N}{j}.\end{aligned}$$

The last equality is the elementary telescoping binomial identity. Replace each t^j in Q_a by the coordinate x_j. The resulting graph has contact through derivative order N with α at a, because $\log t - Q_a(t)$ and its first N derivatives vanish there. By uniqueness it is O_a. □

3 Interpolation at reciprocal nodes

Proof of Theorem 1. Set $r_i = 1/a_i$ and $c_j = (-1)^{j-1}\binom{N}{j}/j$. Define a polynomial of degree at most N by

$$R(r) = z - \sum_{j=1}^{N} c_j x_j r^j.$$

All the c_j are nonzero, so specifying its coefficients is equivalent to specifying $(x_1, \dots, x_N, z)$. Lemma 1 says that membership in all the O_{a_i} is equivalent to

$$R(r_i) = \log a_i - H_N, \qquad 1 \le i \le n.$$

The $n = N+1$ nodes r_i are distinct. Ordinary polynomial interpolation therefore gives exactly one such R, proving existence and uniqueness of the intersection, not merely a formula conditional on its existence.

Its constant coefficient is obtained by evaluating the Lagrange formula at zero:

$$z = R(0) = \sum_{i=1}^{n} (\log a_i - H_N) \prod_{k \ne i} \frac{-r_k}{r_i - r_k}.$$

Each product is

$$\prod_{k \ne i} \frac{-1/a_k}{1/a_i - 1/a_k} = \prod_{k \ne i} \frac{a_i}{a_i - a_k} = w_i.$$

The Lagrange basis polynomials sum to one, so $\sum_i w_i = 1$. Consequently $z = \sum_i w_i \log a_i - H_N$, as claimed. Taking the inverse of the last coordinate function, namely exponentiation, proves the identity of means. □

4 Internality and coinciding inputs

Although the conjecture assumes distinct inputs, the following also checks that the constructed value is a mean and has the expected continuous extension.

Corollary 1. *For all positive inputs, including repetitions, the continuous extension is*

$$M_n(a) = \exp\left(\mathbb{E}\log\Big(\sum_{i=1}^{n} W_i a_i\Big)\right),$$

where W is uniform on the $(n-1)$-simplex. It is symmetric, continuous, strictly increasing in each coordinate, and homogeneous of degree one. It equals a on $(a,\dots,a)$ and lies strictly between the smallest and largest input for every nonconstant tuple.

Proof. For $f(t) = t^N \log t$, the sum in Theorem 1 is the divided difference $[f](a_1,\dots,a_n)$. Repeated differentiation gives

$$f^{(N)}(t) = N!(\log t + H_N).$$

The Hermite–Genocchi formula, obtained by repeatedly applying the fundamental theorem of calculus to divided differences, is

$$[f](a_1,\dots,a_n) = \frac{1}{N!}\mathbb{E}f^{(N)}\Big(\sum_i W_i a_i\Big).$$

This proves the integral formula for distinct inputs. Its right side is continuous for all positive inputs and therefore gives the unique continuous extension. Permutation invariance and scaling follow directly. Since $W_i > 0$ almost surely, increasing any one input increases the logarithm almost surely. For a nonconstant tuple the convex combination lies strictly between its extreme inputs almost surely, proving strict internality. □

The assertion about repeated inputs concerns the continuous mean. Identical hyperplanes by themselves do not determine a unique intersection point.

5 Follow-up tasks 2–5

Task 2. Replace the last coordinate $\log t$ by t^p, with real $p \notin \{0, 1, \dots, n-1\}$. Determine the intersection formula and classify the parameters for which its last coordinate defines a strict mean after taking the inverse coordinate function. Recover the logarithmic case through the limit of $(t^p - 1)/p$ as $p \to 0$.

Task 3. Construct a geometric analogue for the weighted mean $\exp(\mathbb{E}\log\sum_i W_i a_i)$ when W has a prescribed Dirichlet distribution. Determine which weights can be realized by confluent osculating conditions and extend the construction beyond integer parameters.

Task 4. Develop an algorithm for the complete intersection point that stays accurate when the input nodes almost coincide. Prove error bounds and compare its conditioning and cost with direct Vandermonde inversion.

Task 5. Determine sharp constants bounding $\log(\bar{a}/M_n(a))$ by multiples of $n^{-1}\sum_i (a_i - \bar{a})^2/\bar{a}^2$ under $\max_i a_i / \min_i a_i \le R$. Identify extremizers and the asymptotics as $R \downarrow 1$, $R \to \infty$, and $n \to \infty$.

References

[1] A. Horwitz, *A logarithmic mean and intersections of osculating hyperplanes in* $\mathbb{R}^n$, Journal of Classical Analysis 4 (2014), 41–61, final conjecture on p. 61. https://doi.org/10.7153/jca-04-02. Preprint: https://arxiv.org/abs/1404.1489.

[2] MathDB 327618, live record retrieved 11 September 2026. https://mathdb.com/p/327618.